Aakash
Medical | IIT-JEE | Foundations

Home Assignment

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FINAL TEST SERIES for NEET-2024

Test -2

MM : 720 **Time : 3 Hrs. 20 Mins.**

Topics covered :

Physics : Laws of Motion, Work, Energy and Power, System of Particles and Rotational Motion
Chemistry : Chemical Bonding and Molecular Structure, Thermodynamics
Botany : Morphology of Flowering Plants, Anatomy of Flowering Plants
Zoology : Structural Organization in Animals, Biomolecules

Instructions :

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
(iii) Use blue/black ballpoint pen only to darken the appropriate circle.
(iv) Mark should be dark and completely fill the circle.
(v) Dark only one circle for each entry.
(vi) Dark the circle in the space provided only.
(vii) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

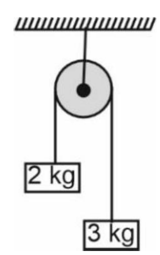
Choose the correct answer :

SECTION - A

1. If a force varies with time as $F = 50 - 10t$, then impulse as a function of time will be

(1) $50t - 5t^2$
(2) $50t - 10$
(3) $10t^2$
(4) $10t^2 + 50$

2. Two bodies of masses 2 kg and 3 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless. The acceleration of system in terms of acceleration due to gravity (g) is

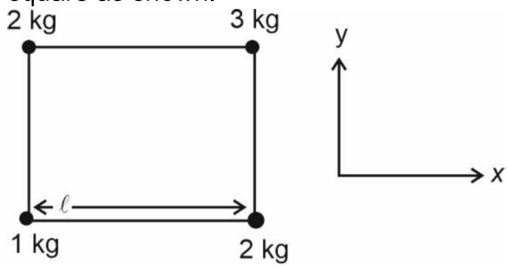


(1) $\frac{g}{2}$
(2) $\frac{g}{5}$
(3) $\frac{g}{7}$
(4) g

3. A body of mass 3 kg moves with an acceleration of 3 m s^{-2}. The change in momentum in one second is

(1) 6 kg m s^{-1}
(2) 9 kg m s^{-1}
(3) 7 kg m s^{-1}
(4) Zero

4. Four-point masses are kept at four corners of a square as shown.



Taking the origin to be at 1 kg mass, then the y-coordinate of COM of system is

(1) $\frac{\ell}{2}$ (2) $\frac{3\ell}{4}$

(3) $\frac{5\ell}{8}$ (4) $\frac{5\ell}{7}$

5. If $\vec{A}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{B}=\hat{i}+2\hat{j}+\hat{k}$, then vector product of $\vec{A}$ and $\vec{B}$ is

(1) $-\hat{i}+\hat{k}$ (2) $\hat{i}-\hat{k}$

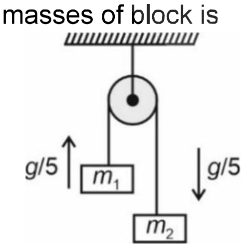
(3) $\hat{i}+\hat{j}-\hat{k}$ (4) $\hat{i}-\hat{j}-\hat{k}$

6. A rocket with lift off mass of 20000 kg is blasted with an initial acceleration of 5 m s^{-2}. The initial thrust of the blast is (g = 10 m s^{-2})

(1) 3×10^5 N (2) 2×10^5 N

(3) 4×10^5 N (4) 5×10^5 N

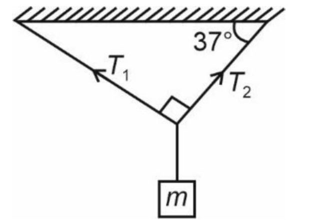
7. A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2. If acceleration of the blocks w.r.t. pulley is $\frac{g}{5}$, then the ratio of masses of block is



(1) $\frac{2}{3}$ (2) $\frac{3}{5}$

(3) $\frac{7}{4}$ (4) $\frac{1}{8}$

8. A point mass m is hanging from two ideal strings as shown below. The ratio of tension T_1 to T_2 is



(1) $\frac{3}{4}$ (2) $\frac{3}{5}$

(3) $\frac{4}{3}$ (4) $\frac{4}{5}$

9. Which of the following statement is correct if net external force acting on the system is zero?

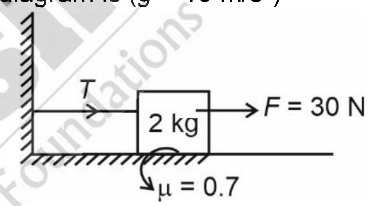
(1) The centre of mass of system must be at rest

(2) The centre of mass of system must accelerate

(3) The centre of mass of system may move with constant velocity

(4) The centre of mass of system must move with constant velocity

10. If the block is at rest, then the tension (T) in the given diagram is (g = 10 m/s^2)



(1) 4 N (2) 21 N

(3) 12 N (4) 16 N

11. A particle of mass 2 kg is moving with velocity 10 m s^{-1} along the straight-line x = 2 m. The magnitude of angular momentum of the particle about the point $P(0, 0)$ will be

(1) 40 kg m^2 s^{-1} (2) Zero

(3) 20 kg m^2 s^{-1} (4) 15 kg m^2 s^{-1}

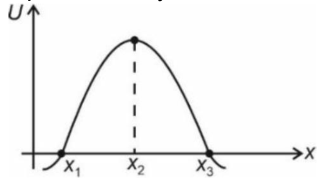
12. If the work done to stretch a spring by x is W, then the additional work done to stretch it further by $2x$ is

(1) $8W$ (2) $3W$

(3) W (4) $2W$

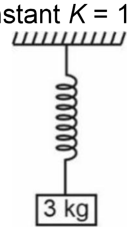
Space for Rough Work

13. The potential energy U of a particle is plotted against its position 'x' from origin. The equilibrium position of particle may be



(1) $x = x_1$ (2) $x = x_2$
(3) $x = x_3$ (4) Both (1) and (3)

14. A block of mass 3 kg is connected to a spring while the spring is in relaxed state. If the block is released, then maximum extension in spring is (Take spring constant K = 1000 N/m)



(1) 1 cm (2) 3 cm
(3) 6 cm (4) 4 cm

15. Which of the given relation is correct? (where symbols have their usual meaning)
(1) $\vec{\tau} = \vec{F} \times \vec{r}$ (2) $\vec{V} = \vec{r} \times \vec{\omega}$
(3) $\vec{V} = \vec{\omega} \times \vec{r}$ (4) $\vec{L} = \vec{p} \times \vec{r}$

16. A particle of mass m is projected with an initial velocity u at an angle θ with horizontal. The work done by gravity during the time it reaches its highest point is
(1) $\frac{1}{2}mu^2\cos^2\theta$ (2) $\frac{1}{2}mu^2\sin^2\theta$
(3) $-\frac{1}{2}mu^2\sin^2\theta$ (4) Zero

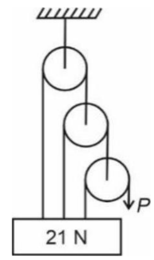
17. The analogous of mass in rotational motion is
(1) Moment of inertia (2) Angular velocity
(3) Radius of gyration (4) Torque

18. The angular momentum of a body is given by equation $L = (3t^2 + 2t + 1)$ kg m^2 s^{-1}. The magnitude of torque acting on body at t = 1 s is
(1) Zero (2) 8 N m
(3) 5 N m (4) 7 N m

19. A particle is moving on the circular path performing uniform circular motion. The power delivered by the force is
(1) Positive (2) Negative
(3) Zero (4) Infinite

20. A particle of mass m is tied to a string of length l and given a circular motion in the vertical plane. If it performs the complete loop, then the difference in tension at the lowest and highest point is
(1) $2mg$ (2) $4mg$
(3) $8mg$ (4) $6mg$

21. The pull P is just sufficient to keep the block of weight 21 N in equilibrium as shown. If the pulleys are assumed to be ideal, then the value of P is



(1) 4 N (2) 6 N
(3) 3 N (4) 2 N

22. A block of mass 1 kg is horizontally thrown with velocity of 10 m/s on a stationary long rough plank whose coefficient of kinetic friction is 0.25. After what time block will come to rest?
(1) 2 s (2) 2.5 s
(3) 4 s (4) 4.3 s

23. When a horse pulls a wagon, the force that causes the horse to move forward is the force
(1) Horse exerts on ground
(2) Horse exerts on wagon
(3) Wagon exerts on horse
(4) Ground exerts on horse

24. A pendulum is hanging from the ceiling of a car having an horizontal acceleration of $\frac{10}{\sqrt{3}}$ m/s^2. What is the angle made by string to vertical? (Take g = 10 ms^{-2})
(1) 30° (2) 37°
(3) 45° (4) 60°

Space for Rough Work

25. A rocket from launching pad is set for vertical lift off. The exhaust speed is 1500 m/s. How much gas must be ejected per second to just lift off the rocket. Mass of rocket is 6000 kg (g = 10 m/s^2)
(1) 112 kg/s
(2) 92 kg/s
(3) 49.5 kg/s
(4) 40 kg/s

26. In the following question, Column-I is matched with Column-II. The incorrect match of columns is

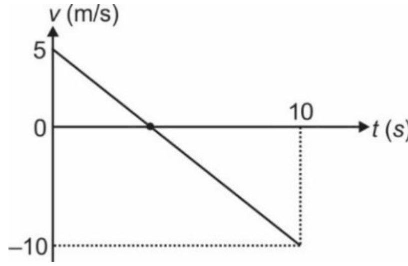
	Column-I		Column-II
A.	Kinetic friction	$\rightarrow$	Independent of area of object
B.	Limiting friction	$\rightarrow$	Maximum value of kinetic friction
C.	Static friction	$\rightarrow$	May be less than limiting friction
D.	Coefficient of kinetic friction	$\rightarrow$	Less than coefficient of static friction

(1) A (2) B
(3) C (4) D

27. If $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = \hat{i} + 2\hat{j}$ then the dot product of both the vector is equal to
(1) 7 unit (2) 11 unit
(3) 5 unit (4) 8 unit

28. **Statement A:** Ability to do the work is called energy.
Statement B: Energy and power is a scalar quantity.
(1) Both A and B are correct
(2) Both A and B are incorrect
(3) A is correct and B is incorrect
(4) A is incorrect and B is correct

29. Velocity (v)-time (t) graph of a particle moving in a straight line is shown in the figure. Mass of the particle is 2 kg. Work done by all the forces acting on the particle in time interval between t = 0 to t =10 s is



(1) 125 J
(2) –125 J
(3) 75 J
(4) –75 J

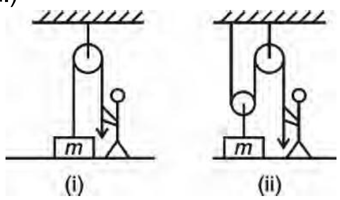
30. A constant force is applied on a 10 kg mass at rest. The ratio of work done in 1st, 2nd and 3rd second will be
(1) 1 : 2 : 3 (2) 1 : 3 : 5
(3) 1 : 4 : 9 (4) 1 : 1 : 1

31. Read the following statements of Assertion (A) and Reason (R) and choose the **correct** option.
Assertion (A): Moment of inertia of two identical sphere having same masses, one solid and other hollow are unequal.
Reason (R): The moment of inertia is minimum when axis of rotation passes through centre of mass.
In the light of above statements, choose the correct option
(1) Both the statements (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both the statements (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true and (R) is false
(4) (A) is false and (R) is true

32. If $\vec{F} = 3\hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{r} = \hat{i} + \frac{2}{3}\hat{j} - \frac{2}{3}\hat{k}$, then the ratio of magnitude of vector product and scalar product of $\vec{F}$ and $\vec{r}$ will be
(1) Zero (2) 1 : 5
(3) $4 : \sqrt{5}$ (4) $\sqrt{5} : 4$

Space for Rough Work

33. A person wants to raise a block lying on the ground to a height h by two ways as shown in the figure. In both cases, if the time required is same, then force exerted by man is (Pulley and strings are ideal)

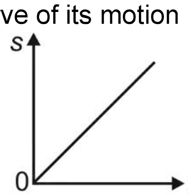


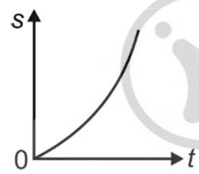
(1) More in case-(i) (2) More in case-(ii)
(3) Same in both cases (4) Data is insufficient

34. Linear velocity of center of mass of a hollow sphere of radius R rotating with angular velocity ω about a fixed axis passing through its center of mass is

(1) $R\omega$ (2) $\frac{R\omega}{2}$
(3) $2R\omega$ (4) Zero

35. A body is moving along a straight line, starting from rest with constant power. The diagram which correctly represent displacement-time (s-t) curve of its motion is

(1)


(2)


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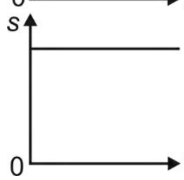
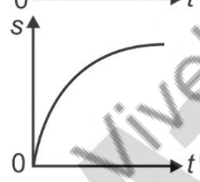
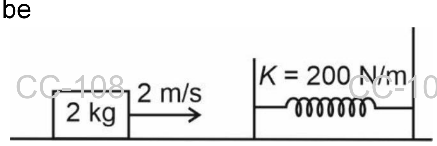


(4)


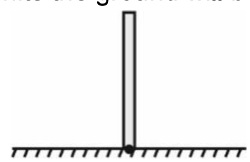
SECTION-B

36. A mass of 2 kg is moving with a speed of 2 m/s on a horizontal smooth surface, collides with a massless spring having spring constant K = 200 N/m. The maximum compression of the spring will be



(1) 10 cm (2) $10\sqrt{3}$ cm
(3) 20 cm (4) $100\sqrt{3}$ cm

37. A slender uniform rod of mass m and length L is pivoted at one end so that it can rotate in vertical plane. The free end is held vertically above the pivot and then released. The angular velocity of rod when it hits the ground will be



(1) $\sqrt{\frac{3g}{L}}$ (2) $\sqrt{\frac{6g}{L}}$
(3) $\sqrt{\frac{5g}{L}}$ (4) Zero

38. The bob of a pendulum at rest is given a horizontal velocity of $\sqrt{7g\ell}$, where ℓ is length of string of pendulum. The tension when bob will reach the highest point of its circular motion is (m is mass of bob)

(1) $2mg$ (2) mg
(3) $4mg$ (4) $7mg$

39. If $\vec{A}\times\vec{B}=\vec{C}$ then $\vec{A}\cdot\vec{C}$ is equal to

(1) AB (2) BC
(3) AC (4) Zero

40. A machine gun fires a bullet of mass 50 g with a velocity of 500 ms^{-1}. The man holding it can exert a maximum force of 150 N on the gun to keep it stationary. The maximum number of bullets that he can fire per second is

(1) 6 (2) 7
(3) 2 (4) 9

41. If $\vec{A}=2\hat{i}+3\hat{j}$ and $\vec{B}=2\hat{j}+3\hat{k}$, then scalar component of $\vec{B}$ along $\vec{A}$ is

(1) 6 (2) $\frac{1}{6}$
(3) $\frac{6}{13}$ (4) $\frac{6}{\sqrt{13}}$

Space for Rough Work

42. Power P (in watt) delivered to a particle, changes with time t (in s) as $P = t^2 + t + 1$. The change in kinetic energy from $t = 0$ to $t = 1$ s is

(1) $\frac{11}{6}$ J (2) 11 J

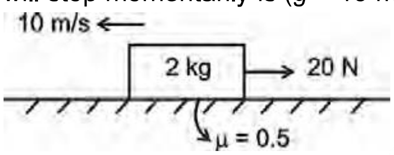
(3) 6 J (4) $\frac{22}{3}$ J

43. A rigid body rotates about a fixed axis with angular speed $\omega = (6 - 3t)$ rad/s where time t is in second. The angle through which it rotates before it comes to rest is

(1) 2 rad (2) 6 rad

(3) 10 rad (4) $\frac{9}{10}$ rad

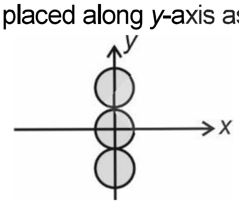
44. A block of mass 2 kg is moving with initial velocity 10 m/s enters on a rough horizontal surface. An extra force of 20 N is opposing the motion and trying to stop the block. The time after which the block will stop momentarily is (g = 10 m/s^2)



(1) 1 s (2) 0.5 s

(3) 0.67 s (4) 0.33 s

45. Three uniform solid spheres each of mass M and radius R are placed along y-axis as shown in figure.



The moment of inertia of system about x-axis is

(1) $\frac{8MR^2}{5}$ (2) $\frac{24MR^2}{5}$

(3) $\frac{46MR^2}{5}$ (4) $\frac{36MR^2}{5}$

46. Mass of a body is the measure of

(1) Velocity of body (2) Acceleration of body

(3) Force on body (4) Inertia of body

47. A particle of mass 4 kg is moving on a circular path of radius 10 m with a speed of 5 m s^{-1} and its speed is increasing at rate 3 m s^{-2}. The magnitude of force acting on particle at this instant is

(1) 7.8 N (2) 11.5 N

(3) 15.6 N (4) 17 N

48. A particle is moving in a circular path of radius a under the action of a conservative force. The potential energy associated with conservative force is $U = \frac{-K}{2r^2}$. Its kinetic energy is given by

(1) $\frac{-K}{2a^2}$ (2) $\frac{K}{2a^2}$

(3) $\frac{-K}{4a^2}$ (4) $\frac{K}{4a^2}$

49. Consider the following statements.

Statement A: If net force on rigid body is zero it must be at rest.

Statement B: Constant speed means acceleration is zero.

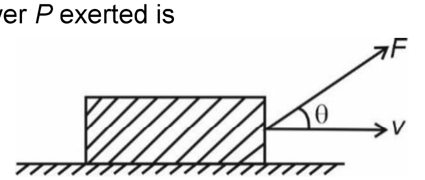
(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct while statement B is incorrect

(4) Statement B is correct while statement A is incorrect

50. A constant force $\vec{F}$ is acting on a body of mass m moving with constant velocity $\vec{v}$ as shown. The power P exerted is



(1) $Fv\cos\theta$ (2) $\frac{F\cos\theta}{mg}$

(3) $\frac{Fmg\sin\theta}{v}$ (4) Fv

Space for Rough Work

CHEMISTRY

SECTION - A

51. Polar and planar molecule among the following is
    (1) BF_3 (2) H_2O
    (3) CH_4 (4) XeO_3
52. Shape of XeF_4 molecule is
    (1) See-saw (2) Tetrahedral
    (3) Square planar (4) Octahedral
53. Zero overlap takes place between which of the following, if x-axis is the inter-nuclear axis?
    (1) 1*s* and 2*s* (2) $2p_x$ and 2*s*
    (3) $2p_x$ and $2p_y$ (4) $2p_x$ and 1*s*
54. In the given compound $CH_3-\overset{*}{C}H=C=CH_2$, hybridisation of the marked carbon atom is
    (1) *sp* (2) sp^2
    (3) sp^3 (4) dsp^2
55. Number of 90° angle(s) in ClF_5 is
    (1) Zero (2) 3
    (3) 5 (4) 6
56. Given below are two statements.
    Statement I: Both NF_3 and NH_3 are pyramidal in shape.
    Statement II: Dipole moment of NF_3 is greater than NH_3.
    In light of the above statements, choose the correct answer.
    (1) Both statement I and statement II are correct
    (2) Both statement I and statement II are incorrect
    (3) Statement I is incorrect but statement II is correct
    (4) Statement I is correct but statement II is incorrect
57. Which of the following molecular orbital contains two nodal planes?
    (1) $\sigma 1s$ (2) σ^*1s
    (3) $\pi^* 2p_x$ (4) $\sigma^* 2p_z$
58. Which of the following species contains only π bond?
    (1) H_2 (2) C_2
    (3) N_2 (4) O_2
59. Paramagnetic species among the following is/are
    I. N_2^+ II. N_2^-
    III. O_2
    (1) I only (2) II & III only
    (3) III only (4) I, II & III
60. Which of the following species has fractional bond order?
    (1) He_2^+ (2) H_2
    (3) Li_2 (4) Be_2
61. Given below are two statements.
    Statement I : On boiling of egg entropy decreases.
    Statement II : On evaporation of water, entropy increases.
    In light of the above statements, choose the correct answer.
    (1) Statement I is correct but statement II is incorrect
    (2) Statement I is incorrect but statement II is correct
    (3) Both statement I and statement II are incorrect
    (4) Both statement I and statement II are correct
62. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R)
    Assertion (A) : On mixing two liquids with T_1 and T_2 temperature respectively, final temperature of mixture will not be $T_1 + T_2$.
    Reason (R) : Temperature is an intensive property.
    In the light of the above statements.
    Choose the correct answer from the options given below:
    (1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
    (2) (A) is true but (R) is false
    (3) (A) is false but (R) is true
    (4) Both (A) and (R) are true and (R) is the correct explanation (A)

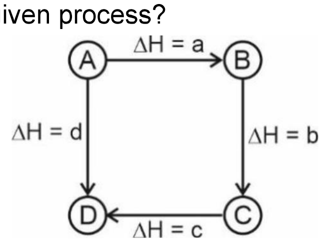
Space for Rough Work

63. Heat capacity for isothermal process is
(1) Zero (2) 1
(3) Infinite (4) R

64. Work done in a reversible isothermal expansion is given by
(1) $2.303\ nRT \log \frac{V_f}{V_i}$ (2) $-2.303\ nRT \log \frac{P_i}{P_f}$
(3) $-2.303\ nR \log \frac{V_f}{V_i}$ (4) $2.303\ nR \log \frac{P_i}{P_f}$

65. The amount of heat required to raise the temperature of 21 g of aluminium from 250 K to 350 K is (Specific heat of aluminium is 0.4 J (°C)$^{-1}$ g^{-1})
(1) 420 J (2) 1100 J
(3) 840 J (4) 2400 J

66. Which among the following is a correct relation for the given process?



(1) a – b = d – c (2) a + c = d – b
(3) b + c = a + d (4) a – d = b + c

67. Given below are two statements.
Statement I : Standard heat of formation for O_3 is zero.
Statement II : O_3 is a naturally occurring molecule.
In light of the above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are incorrect
(4) Both statement I and statement II are correct

68. The third law of thermodynamics deals with
(1) Absolute entropy
(2) Enthalpy
(3) Free energy
(4) Internal energy

69. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R)
Assertion (A) : ΔH is negative for irreversible free expansion of an ideal gas under isothermal condition.
Reason (R) : Irreversible free expansion of an ideal gas is spontaneous process.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(2) (A) is true but (R) is false
(3) (A) is false but (R) is true
(4) Both (A) and (R) are true and (R) is the correct explanation (A)

70. ΔH_f for C_2H_4 = 12.5 kcal mol^{-1}
Heat of atomisation of C = 171 kcal mol^{-1}
Bond energy of H_2 = 104.3 kcal mol^{-1}
Bond energy C – H = 99.3 kcal mol^{-1}
Bond energy of C = C will be
(1) 140.9 kcal mol^{-1} (2) 76 kcal mol^{-1}
(3) 86 kcal mol^{-1} (4) 41 kcal mol^{-1}

71. If calorific value of CH_4 is x kJ/g then the heat of combustion of CH_4 (in kJ/mol) is
(1) $\frac{x}{16}$ (2) 16x
(3) x (4) $\frac{16}{x}$

72. The value of ΔH and ΔS for the reaction, C(graphite) + CO_2(g) $\rightarrow$ 2CO(g) are 170 kJ mol^{-1} and 170 J K^{-1} mol^{-1} respectively. This reaction will be in equilibrium at
(1) 950 K (2) 900 K
(3) 1100 K (4) 1000 K

Space for Rough Work

73. Match the species given in List I with their shapes given in List II and choose correct answer

	List I		List II
a.	SF_6	(i)	Tetrahedral
b.	CCl_4	(ii)	Trigonal planar
c.	ClF_3	(iii)	Octahedral
d.	SO_3	(iv)	T-shape

	a	b	c	d
(1)	(ii)	(iii)	(i)	(iv)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(i)	(ii)	(iv)	(iii)
(4)	(iv)	(iii)	(i)	(ii)

74. The hybridisation of I in ICl_3 is

(1) sp^3d (2) sp^3d^2

(3) sp^3 (4) dsp^2

75. Formal charge on N in $\ddot{O}=\dot{N}-\ddot{O}:$ is

(1) Zero (2) –1

(3) +1 (4) +2

76. Which among the given molecules does not contain central atom with expanded octet?

(1) PF_5 (2) SF_6

(3) H_2SO_4 (4) H_2O

77. Given below are two statements.

Statement I: Acetone is soluble in water.

Statement II: Acetone forms intermolecular hydrogen bond with water.

In light of the above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

78. Match the species in List I with their bond order according to MOT in List II and choose the correct answer.

List I (Species)		List II (Bond Order)	
a.	O_2^+	(i)	2
b.	C_2	(ii)	2.5
c.	N_2	(iii)	1
d.	F_2	(iv)	3

(1) a(i); b(ii); c(iv); d(iii)

(2) a(ii); b(i); c(iv); d(iii)

(3) a(ii); b(i); c(iii); d(iv)

(4) a(i); b(iv); c(ii); d(iii)

79. For one mole of a gas, γ is equal to 1.4, the gas will be

(1) Monoatomic (2) Diatomic

(3) Triatomic (4) Tetraatomic

80. Necessary condition for spontaneity is

(1) $\Delta S_{system} = 0$ (2) $\Delta H_{system} < 0$

(3) $\Delta G = 0$ (4) $\Delta S_{total} > 0$

81. Consider the following reactions

(i) C (graphite, s) + $2H_2O(l) \rightarrow CO_2(g) + 2H_2(g)$; $\Delta H° = -a$ kJ/mol

(ii) C (graphite, s) + $O_2(g) \rightarrow CO_2(g)$; $\Delta H° = -b$ kJ/mol

(iii) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$; $\Delta H° = -c$ kJ/mol

(iv) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$; $\Delta H° = -d$ kJ/mol

$\Delta_f H°$ of CO_2 (g) will be

(1) – a kJ/mol (2) – b kJ/mol

(3) – c kJ/mol (4) – d kJ/mol

82. If enthalpies of formation of $CH_4(g)$, $CO_2(g)$ and $H_2O(l)$ at 25°C and 1 atm pressure are –x, –y and – z kJ/mol respectively, then enthalpy of combustion of CH_4 (g) will be

(1) x – y – z kJ/mol

(2) x – y – 2z kJ/mol

(3) z + x – y kJ/mol

(4) z – 2x – y kJ/mol

Space for Rough Work

83. The change in internal energy if 100 J of heat is given to the system at constant pressure and 30 J of work is done by the system will be
(1) – 130 J (2) 130 J
(3) – 70 J (4) 70 J

84. Which among the following acid will give maximum heat of neutralisation with NaOH?
(1) HCl (2) CH_3COOH
(3) HCOOH (4) HF

85. Which among the following has equal number of σ and π bonds?
(1) C_2H_2 (2) $C_2(CN)_4$
(3) C_6H_{12} (4) C_2H_4

SECTION - B

86. Which of the following species has maximum bond angle?
(1) H_2O (2) NH_3
(3) CH_4 (4) BF_3

87. For which of the following reactions, $\Delta H = \Delta U$?
(1) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
(2) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(3) $2HI(g) \rightleftharpoons I_2(g) + H_2(g)$
(4) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

88. For the following processes

			ΔH (kJ/mol)
A	$\rightarrow$	B	– x
2B	$\rightarrow$	C + D	– y
E + B	$\rightarrow$	3C	– z

For, $2A + 3B \rightarrow 3D + E$, ΔH in kJ mol^{-1} will be
(1) 2z + x – 2y (2) z – 2x + 3y
(3) –z + 2x – 2y (4) z – 2x – 3y

89. Which of the following overlapping will result into strongest bond formation?
(1) $2s - 2s$ (2) $2p - 2p$ axial
(3) $3s - 3s$ (4) $2s - 2p$ axial

90. If enthalpies of neutralisation of four acids A, B, C and D are –50, –20, –53 and –42 kJ eq^{-1}, respectively when they are neutralised by a common base then the strongest acid among the following is
(1) A (2) B
(3) C (4) D

91. Entropy of vaporisation of water at 100°C, if molar heat of vaporisation is 7460 cal mol^{-1} will be
(1) 25 cal K^{-1} mol^{-1}
(2) 20 cal K^{-1} mol^{-1}
(3) 30 cal K^{-1} mol^{-1}
(4) 28 cal K^{-1} mol^{-1}

92. Lowest unoccupied molecular orbital for F_2 molecule is,
(1) π^*2p_x (2) σ^*2p_z
(3) $\pi2p_x$ (4) $\sigma2p_z$

93. If the heat of combustion of C (graphite) is –400 kJ/mol, then the heat released upon the formation of 88 g of CO_2 from C (graphite) is
(1) 800 kJ (2) 300 kJ
(3) 600 kJ (4) 1500 kJ

94. The work done during the expansion of a gas from a volume of 2 L to 6 L against a constant external pressure of 5 atm is
(1) Zero (2) –5 L atm
(3) –20 L atm (4) –10 L atm

95. Hydrogen bonding is not present in
(1) H_2O (2) CH_3OH
(3) HF (4) HCl

96 Compound which is most easily soluble in water among the following is
(1) CCl_4 (2) C_6H_6
(3) C_2H_5OH (4) C_4H_{10}

97. If bond energy of 'P – P', 'H – H' and P – H bonds are akJ mol^{-1}, bkJ mol^{-1} and ckJ mol^{-1} respectively the heat of following reaction (in kJ mol^{-1}) will be,
$P_4(g) + 6H_2(g) \rightarrow 4PH_3(g)$
(1) 4a + 6b – 12c
(2) 12c + 4a – 6b
(3) 12c + 6a – 6b
(4) 6a + 6b – 12c

98. Correct order of bond length for the given species is
(1) C – O > C – C > O – H > C – H
(2) C – C > C – O > O – H > C – H
(3) C – O > C – C > C – H > O – H
(4) C – C > C – O > C – H > O – H

Space for Rough Work

99. A gas is allowed to expand in a well-insulated container against a constant external pressure of 4.0 atm from an initial volume of 1.5 L to a final volume of 7.5 L. The change in internal energy (ΔU) of the gas will be approximately.

(1) –33.1 kJ (2) –2.4 kJ
(3) –4.7 kJ (4) –1.2 kJ

100. Which among the following does not contain $p\pi - d\pi$ bond?

(1) CO_3^{2-}
(2) SO_4^{2-}
(3) PO_4^{3-}
(4) ClO_4^-

BOTANY

SECTION - A

101. Pith and pericycle is generally found to be absent in

(1) Dicot root (2) Dicot stem
(3) Monocot root (4) Monocot stem

102. All of the following are present in gymnosperms **except**

(1) Albuminous cells (2) Sieve cells
(3) Vessels (4) Tracheids

103. ________ shows whorled phyllotaxy.

Select the **correct** option to fill in the blank.

(1) China rose (2) Guava
(3) *Alstonia* (4) *Calotropis*

104. Cypsela is the fruit of which of the following family?

(1) Compositae (2) Cruciferae
(3) Fabaceae (4) Poaceae

105. Read the following statements and choose the **correct** option.

Statement I: Many of the cells of epiblema protrude in the form of multicellular root hairs.

Statement II : All tissues exterior to the vascular cambium constitute the stele.

(1) Both statements I and II are correct
(2) Only statement I is correct
(3) Only statement II is correct
(4) Both statements I and II are incorrect

106. Select the **correctly** matched pair.

(1) Cork cambium – Develops usually in cortex region
(2) Heartwood – Involved in conduction of water and minerals
(3) Lenticels – Least occur in woody trees
(4) Phelloderm – Impervious to water due to suberin deposition

107. In cymose inflorescence

(a) Main axis terminates into a flower
(b) Flowers are borne in an acropetal succession
(c) Unlimited growth of main axis occurs

Choose the **correct** ones.

(1) a and b only (2) b only
(3) a only (4) b and c only

108. Match the aestivation of petals given in Column I with plants given in Column II.

Column I	Column II
a. Imbricate aestivation	(i) *Calotropis*
b. Twisted aestivation	(ii) Pea
c. Vexillary aestivation	(iii) Gulmohur
d. Valvate aestivation	(iv) Cotton

Choose the **correct** answer from the options given below.

(1) a(ii), b(iv), c(iii), d(i)
(2) a(iii), b(iv), c(ii), d(i)
(3) a(ii), b(iii), c(i), d(iv)
(4) a(i), b(ii), c(iii), d(iv)

Space for Rough Work

109. Mango and coconut are similar in
(1) Having fleshy edible mesocarp
(2) Being developed from monocarpellary superior ovaries
(3) Not being one seeded
(4) Having fibrous endocarp

110. State **T (True)** OR **F (False)** to the given statements and choose the option accordingly.
(A) When bulliform cells are flaccid, they make the leaves curl inwards.
(B) The conjoint vascular bundles usually have the xylem located only on the outer side of phloem.
(C) All tissues except epidermis constitute the ground tissue which forms main bulk of plant.
(D) In winter, cambium is very active and forms large number of xylary elements.

	(A)	(B)	(C)	(D)
(1)	F	T	T	F
(2)	T	F	T	T
(3)	T	F	F	F
(4)	F	T	T	T

111. Regarding various regions of the root-tip, find the **mismatched** pair.

(1)	Root cap	–	Protects the tender apex of the root
(2)	Region of elongation	–	Responsible for growth of root in length
(3)	Region of maturation	–	Presence of root hairs
(4)	Region of meristematic activity	–	Covers the root cap and protects it

112. Select the **correct** statement w.r.t. vessels.
(1) It is a long, cylindrical tube-like structures made up of many cells called vessel members
(2) Its cells are thin with cellulosic cell wall and without lignin
(3) They are less efficient elements of xylem for conduction of water due to not forming continuous column.
(4) Its cells are rich in protoplasm.

113. Which of the following features is said **not** to be true for phylloclade?
(1) Shows unlimited growth
(2) Green in colour and photosynthetic in nature
(3) Modified fleshy cylindrical structure
(4) Has large number of stem thorns

114. There are usually two to four xylem and phloem patches in
(1) Monocot roots (2) Monocot stems
(3) Dicot roots (4) Dicot stems

115. In vexillary aestivation, the largest petal that overlaps the two smaller lateral petals is called
(1) Keel (2) Wings
(3) Standard (4) Pedicel

116. Choose the **false** statements regarding sieve tube elements.
(1) These are thin walled living structures
(2) They have peripheral cytoplasm with prominent nucleus
(3) These are food conducting elements of plants
(4) Sieve tube and companion cells are sister cells

117. How many of the following plants belong to Solanaceae family and are mainly used as medicinal plant?

Aloe, Belladonna, Lupin, Sweet pea, *Petunia, Muliathi*, *Gloriosa, Ashwagandha*

(1) Two (2) Five
(3) Six (4) Four

118. Choose the **correct** option to fill the blank.
'In grasses, the guard cells are ____ .
(1) Round
(2) Bean shaped
(3) Kidney shaped
(4) Dumb-bell shaped

119. Aleurone layer in monocotyledonous seeds is
(1) Triploid and is part of scutellum
(2) Diploid and is part of endosperm
(3) Diploid and is part of perisperm
(4) Triploid and is outer covering of endosperm

Space for Rough Work

120. Vascular cambium of dicot roots
(1) Is primary in origin and circular in outline
(2) Is partly primary and partly secondary
(3) Is formed due to redifferentiation
(4) Is completely secondary in origin and form a wavy ring initially

121. In sunflower root, vascular bundles and arrangement of primary xylem respectively are
(1) Conjoint - open and Endarch
(2) Conjoint - closed and Exarch
(3) Radial and Exarch
(4) Radial and Endarch

122. Meristem which occurs at the tip of the plant and occupies the distant most region of stem axis is
(1) Intercalary meristem
(2) Shoot apical meristem
(3) Lateral meristem
(4) Secondary meristem

123. Which of the following is **not** a modifications of stem?
(1) Spines in cacti
(2) Tendrils in cucumber
(3) Thorns in *Citrus*
(4) Offset in *Eichhornia*

124. Large shield shaped cotyledon in maize is known as
(1) Hilum (2) Scutellum
(3) Coleoptile (4) Coleorhiza

125. In lemon, which type of arrangement of ovules within the ovary is found?
(1) Marginal (2) Axile
(3) Free central (4) Parietal

126. Dicot leaf differs from monocot leaf as the former
(a) Has bulliform cells
(b) Has usually more number of stomata on lower epidermis in comparison to the upper epidermis
(c) Mesophyll is not differentiated into palisade and spongy parenchyma
(d) Has vascular bundles which differ in size due to reticulate venation.

The **correct** one(s) is/are
(1) a and c only (2) a, b and c only
(3) b and d only (4) d only

127. The permanent tissue, collenchyma
(1) Occurs in layers below epidermis in most of monocots
(2) Forms major component within various organs of plants
(3) Is composed of cells that has thickenings of cellulose, hemicellulose and pectin at the corners of their cell wall
(4) Has large intercellular spaces

128. Select the **incorrect** statement.
(1) Pneumatophores are modified roots which help to get oxygen for respiration
(2) In *Monstera,* roots arise from parts of the plants other than radicle
(3) In Australian acacia, leaves are large and long-lived
(4) In monocotyledons, leaf base expands into a sheath covering stem partially or wholly

129. Members of lily family have all of the below given features, **except**
(1) Leaves are exstipulated with parallel venation
(2) Presence of tricarpellary, apocarpous, superior ovary
(3) Perennial herbs with underground rhizomes or corms
(4) Stamens are six in number with epitepalous condition

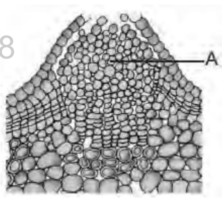
130. Read the following statements and select the **correct** option.
Assertion (A) : Generally monocotyledonous seeds are endospermic as in orchids.
Reason (R) : In orchids, endosperm is not completely consumed during embryo development.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(2) Both (A) and (R) are true and (R) is the correct explanation of (A).
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

Space for Rough Work

131. Which of the following is **not** a lateral meristem?
(1) Cork cambium
(2) Intercalary meristem
(3) Fascicular vascular cambium
(4) Interfascicular cambium

132. The modified underground stems can perform all the given functions, **except**
(1) Organs of perennation
(2) Storing food materials
(3) Vegetative reproduction
(4) Photosynthesis

133. Select the **correct** feature for the cells labelled as A.



(1) Thin walled cells
(2) Suberized cell walls
(3) Meristematic cells
(4) Dedifferentiated cells

134. Periderm includes
(1) Phellogen and phellem only
(2) Phellogen, phellem and phelloderm
(3) Phellem and phelloderm only
(4) Primary cortex and pericycle only

135. Swollen placenta with large number of ovules is the characteristic of a family whose members also
(1) Have umbellate clusters of flower
(2) Shows fibrous and adventitious roots
(3) Have ex-albuminous seeds
(4) Have persistent calyx

SECTION - B

136. Storage structure in sweet potato is a modified
(1) Leaf (2) Stem
(3) Root (4) Axillary bud

137. ________ is/are commonly found in seed coats of legumes and leaves of tea and provide(s) hardness to the structures.
Select the **correct** option to fill in the blank.
(1) Parenchyma (2) Sclereids
(3) Collenchyma (4) Chlorenchyma

138. Read the following statements and choose the **correct** option for them.
Statement I: In seeds of cereals, seed coat is membranous and generally fused with fruit wall.
Statement II: In *Salvia*, there may be variation in the length of stamen filaments within a flower.
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

139. In dicot stem
(1) Hypodermis is sclerenchymatous
(2) Ring arrangement of vascular bundles is observed
(3) There is absence of starch sheath
(4) Water containing cavities are present in vascular bundles

140. Read the following statements and select the **correct** option.
Assertion (A) : Secondary growth also occurs in the stems and roots of grasses.
Reason (R) : Vascular bundles of grasses have cambium which is present between phloem and xylem.
(1) Both (A) and (R) are false
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(4) (A) is true but (R) is false

141. All are related to guard cells, **except**
(1) In dicots, inner walls are thick
(2) These are sclerenchymatous
(3) Contain chloroplast
(4) Regulate opening and closing of stomata

142. Modified adventitious roots for mechanical support in sugarcane and banyan are respectively called
(1) Prop roots and assimilatory roots
(2) Stilt roots and prop roots
(3) Fibrous roots and stilt roots
(4) Prop roots and respiratory roots

Space for Rough Work

143. Tyloses are
   (1) Balloon-like outgrowths of tracheid cells in lumen of vessels
   (2) Present in sieve tubes
   (3) Temporary thickenings of tracheid walls formed in monocots in winters
   (4) Present in heart wood and make it non-functional for water conduction

144. The main function of stem in most of the plants is
   (1) Conduction of water and minerals from leaves to roots
   (2) To provide anchorage to the plant
   (3) Spreading out branches bearing leaves, flowers and fruits
   (4) Vegetative propagation

145. Which of the following is **not** formed by the process of redifferentiation?
   (1) Phellogen (2) Phellem
   (3) Phelloderm (4) Secondary xylem

146. Choose the **correct** match.

(1)	Primary growth	–	By lateral meristem
(2)	Secondary growth	–	By apical meristem
(3)	Increase in girth	–	Due to vascular cambium and cork cambium
(4)	Medullary ray cells	–	Are intrafascicular cambium

147. Which of the following features are **correct** w.r.t. *Indigofera.*
   a. Superior ovary
   b. Diadelphous condition of stamens
   c. Actinomorphic flower
   d. Gynoecium monocarpellary
   e. Cymose inflorescence
   (1) All except (d) (2) (a), (b) and (d) only
   (3) (a), (c) and (e) only (4) (b) and (e) only

148. Select the **correct** statement about pericycle.
   (1) It is always composed of cells that are thick walled and sclerenchymatous
   (2) It is outermost portion of stele
   (3) In monocot roots, it forms cambial ring
   (4) It gives rise to lateral roots in monocots only

149. $\oplus$ ⚥ $K_{(5)} \, \overset{\frown}{C_{(5)} A_5} \, \underline{G}_{(2)}$ is the floral formula of family
   (1) Liliaceae (2) Solanaceae
   (3) Brassicaceae (4) Fabaceae

150. Which of the following tissues is innermost in a woody stem showing secondary growth?
   (1) Primary xylem (2) Secondary xylem
   (3) Secondary phloem (4) Primary phloem

ZOOLOGY

SECTION - A

151. The positional information of amino acids is given by which of the following structures of a protein?
   (1) Primary
   (2) Secondary
   (3) Tertiary
   (4) Quaternary

152. Select the **odd** one w.r.t heteropolymers.
   (1) RNA
   (2) Insulin
   (3) Cellulose
   (4) Glucagon

153. Deoxyribonucleotides do not include
   (1) Deoxyguanylic acid
   (2) Deoxycytidine
   (3) Uridylic acid
   (4) Deoxyribose + adenine + phosphate

154. Select the molecule that is not formed by dehydration process.
   (1) Collagen
   (2) Guanine
   (3) Triglyceride
   (4) Inulin

Space for Rough Work

155. At each step of ascent in the B-DNA molecule, the strand turns
(1) 72º (2) 36º
(3) 34º (4) 10º

156. Zinc acts as a co-factor for a proteolytic enzyme. It forms __A__ bonds with side chains at active site and at the same time forms one or more __B__ bonds with the substrate. Choose the option which fill the blanks A and B **correctly**.

	A	B
(1)	Coordination	Hydrogen
(2)	Peptide	Coordination
(3)	Covalent	Ester
(4)	Coordination	Coordination

157. Complete the **correct** sequence with the appropriate option w.r.t. digestive system of cockroaches.
Mouth $\rightarrow$ X $\rightarrow$ oesophagus $\rightarrow$ Y $\rightarrow$ proventriculus
(1) X is gizzard; Y is pharynx
(2) X is pharynx; Y is crop
(3) X is hepatic caeca; Y is ileum
(4) X is mesenteron; Y is hepatic caeca

158. How many amino acids and peptide bonds will be present in a hexapeptide containing a cyclic amino acid respectively?
(1) 5 and 6 (2) 6 and 6
(3) 6 and 5 (4) 6 and 7

159. The bond present between two successive amino acids for the formation of insulin can be cleaved with the help of ______ enzyme.
Choose the **correct** option to fill in the blank.
(1) Class II (2) Class IV
(3) Class VI (4) Class III

160. How many phosphodiester bonds are present in 5'AGACTAATG3' sequence of nucleotides?
(1) Nine (2) Seven
(3) Eight (4) Ten

161. Among the given elements, which one makes up maximum percentage of weight of the human body?
(1) Oxygen (2) Hydrogen
(3) Carbon (4) Nitrogen

162. A drug extracted from the latex of *Papaver somniferum* is given to patients as an effective sedative and painkiller.
Select the compound which comes under the same category of the secondary metabolite as the above mentioned drug.
(1) Rubber (2) Anthocyanin
(3) Codeine (4) Gums

163. **Assertion (A) :** The structure of an amino acid changes according to the pH of a solution.
Reason (R) : The ionization state of $-NH_2$ and $-COOH$ group present in the amino acid is influenced by the pH of the surrounding environment due to ionisable nature of $-NH_2$ and $-COOH$ groups.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true and (R) is false
(4) Both (A) and (R) are false

164. In a chain of glycogen, the right end is called the (A) end and the left end is called the (B) end.
Select the **correct** option for (A) and (B) respectively.
(1) Reducing, non-reducing
(2) N-terminal, C-terminal
(3) Non-reducing, reducing
(4) C-terminal, N-terminal

165. Choose the **incorrect** match.
(1) Thymidine – Nitrogenous base + Pentose sugar
(2) Palmitic acid – A saturated fatty acid with 20 carbons
(3) Uracil – A heterocyclic compound
(4) Starch – A homopolysaccharide that acts as the store house of energy in plants

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166. Read the statements given below carefully.
   (a) Trypsin and ribozyme are proteinaceous in nature.
   (b) Competitive inhibitors are often used in the control of bacterial pathogens.
   (c) Activity of enzymes increases both below and above the optimum temperature and pH range.
   (d) Rate of a physical or chemical process refers to the amount of product formed per unit time.

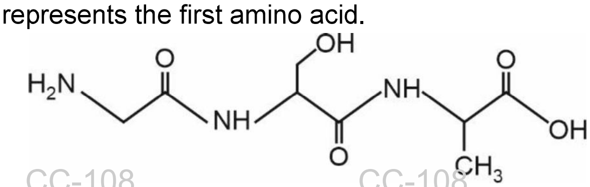
   Select the option with the **correct** statements only.
   (1) (a) and (b)
   (2) (b) and (c)
   (3) (c) and (d)
   (4) (b) and (d)

167. Co-enzymes differ from prosthetic groups in the fact that
   (1) They require metal ions for their activity.
   (2) Their removal does not affect the catalytic activity of the enzyme.
   (3) They are tightly bound to the apoenzymes.
   (4) Their association with apoenzymes is transient.

168. Read the following statements and choose the option with all the **correct** statements.
   a. α-amino acids are substituted methanes.
   b. Amino acids are classified only on the basis of number of $-NH_2$ group present in their structures.
   c. There are only 20 amino acids present in different protein molecules.
   d. Hydroxy methyl is the functional 'R-group' in alanine.
   (1) a and b
   (2) b and c
   (3) c and d
   (4) a and c

169. Arrange the steps of catalytic action of an enzyme in order and choose the **correct** option.
   (i) The enzyme releases the products of the reaction and becomes free.
   (ii) The active site of enzyme is in close proximity of the substrate and breaks the chemical bonds of the substrate.
   (iii) The binding of the substrate induces the enzyme to alter its shape.
   (iv) The substrate binds to the active sites of the enzyme.
   (1) (iv)$\rightarrow$(iii) $\rightarrow$ (ii) $\rightarrow$ (i)
   (2) (ii) $\rightarrow$ (iv) $\rightarrow$ (iii) $\rightarrow$ (i)
   (3) (i) $\rightarrow$ (iv) $\rightarrow$ (iii) $\rightarrow$ (ii)
   (4) (iii) $\rightarrow$ (ii) $\rightarrow$ (i) $\rightarrow$ (iv)

170. Observe the primary structure of a hypothetical protein given below and select the option which represents the first amino acid.



   (1) Valine
   (2) Alanine
   (3) Serine
   (4) Glycine

171. The type of muscles that are found in the walls of blood vessels have muscle fibres that
   (1) Are multinucleated
   (2) Are cylindrical in shape
   (3) Show distinct alternate bands
   (4) Are unbranched

172. Read the statements given below and choose the **correct** option.
   Statement A : All voluntary muscles are striated but all striated muscles are not voluntary.
   Statement B : All smooth muscles are involuntary but all involuntary muscles are not smooth.
   (1) Both statements A and B are correct
   (2) Both statements A and B are incorrect
   (3) Only statement A is incorrect
   (4) Only statement B is incorrect

173. Read the following statements carefully w.r.t connective tissues.
   (1) It is the most abundant and widely distributed tissue in the body of complex animals.
   (2) Adipose tissue is a reservoir of stored energy.
   (3) The fibres provide strength, flexibility, elasticity and conductivity to the tissues.
   (4) In all connective tissues, the cells secrete matrix and fibres of structural proteins.

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How many of the above-mentioned statement(s) are correct?

(1) Two (2) One
(3) Four (4) Three

174. Select the **correct** option to complete the analogy.

Intercalated discs: Cardiac muscle :: Multinucleated fibres: ________

(1) Stomach
(2) Intestine
(3) Urinary bladder
(4) Biceps

175. If a scientist wants to study about junctions that stop leakage of substances across tissues, he must take the study sample from

(1) Kidney tubules
(2) Fluid connective tissues
(3) Thigh muscles
(4) Cranial nerves

176. Select the **odd** one w.r.t sense organs in frogs that are not present as well-organised structures.

(1) Organ of equilibrium
(2) Organ of smell
(3) Organ of touch
(4) Organ of taste

177. In frogs, a triangular structure called __A__ joins the __B__ and receives blood through the major veins called __C__.

Select the option to correctly fill the blanks A, B and C respectively.

	A	B	C
(1)	Sinus venosus	Left atrium	Vena cava
(2)	Conus arteriosus	Right atrium	Aorta
(3)	Sinus venosus	Right atrium	Vena cava
(4)	Conus arteriosus	Left atrium	Aorta

178. Choose the **incorrect** statement w.r.t adult cockroaches.

(1) Their exoskeleton is divided into segments, each containing a dorsal tergite and a ventral sternite.
(2) The pronotum is located on the dorsal surface of the first thoracic segment.
(3) Their head is formed by the fusion of 10 segments.
(4) They have chewing or biting type of mouthparts.

179. Match column I and column II w.r.t abdominal segments of *Periplaneta americana*.

	Column I		Column II
a.	Anal cerci	(i)	4^{th}-6^{th} segments
b.	Testes	(ii)	2^{nd} -6^{th} segments
c.	Spermatheca	(iii)	10^{th} segment
d.	Ovaries	(iv)	6^{th} segment

Choose the **correct** option.

(1) a(iii), b(iv), c(i), d(ii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(i), b(iii), c(ii), d(iv) (4) a(iii), b(i), c(iv), d(ii)

180. Select the **incorrect** option w.r.t frogs.

(1) They are poikilotherms.
(2) The body is divisible into head, neck and trunk.
(3) Their eyes are bulged and situated in an orbit in the skull.
(4) They show protective coloration as defence mechanism.

181. The final site for digestion of food in adult frogs is

(1) Stomach (2) Rectum
(3) Intestine (4) Cloaca

182. Choose the **correct** option to complete the analogy w.r.t type of blood circulation.

Homo sapiens: Double circulation : : *Rana tigrina* : ____

(1) Incomplete single circulation
(2) Incomplete double circulation
(3) Double circulation
(4) Open circulation

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183. Cockroaches differ from humans in all of the following features, **except**

(1) Type of blood vascular system

(2) Type of excretory product

(3) Type of vision

(4) Mode of fertilisation

184. The external changes which are visible after the last moult of a cockroach nymph is

(1) Labrum develops

(2) Development of anal cerci

(3) Both forewings and hindwings develop from wing pads

(4) Formation of alimentary canal

185. Which of the following is **correct** w.r.t. respiratory system of cockroach?

(1) It consists of tracheoles which are divided into tracheal tubes.

(2) 12 spiracles are present on the lateral side of the body.

(3) The opening of the spiracles is not regulated by sphincters.

(4) Exchange of gases takes place at the tracheoles by diffusion.

SECTION - B

186. 'Ramachandran plot' is associated with

(1) Secondary structure of proteins

(2) Ball and stick model of DNA

(3) Linear structure of cellulose

(4) Branched structure of fatty acids

187. Read the statements given below and select the **correct** option.

Statement I : Haemoglobin exhibits quaternary level of structural organisation.

Statement II : An adult human haemoglobin consist of 4 subunits, two of these are identical to each other.

(1) Both statements I and II are incorrect

(2) Only statement I is correct

(3) Only statement II is correct

(4) Both statements I and II are correct

188. How many nitrogen atom(s) is/are present in the structure of lecithin? Select the **correct** option.

(1) Four (2) Three

(3) Two (4) One

189. Select the more likely reason for the presence of lipids in acid-insoluble pool during the chemical analysis of living tissues.

(1) They have very high molecular weight.

(2) They are polymers of glycerol and fatty acids.

(3) They have very high density.

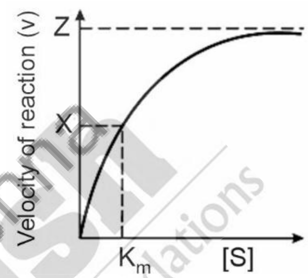
(4) On grinding, the cell membranes and other membranes are broken into pieces and form water insoluble vesicles.

190. A double stranded B-DNA measures 340 nm in length. How many base pairs are possible in this structure?

(1) 340 (2) 680

(3) 960 (4) 1000

191.



Observe the figure given above and select the **incorrect** option w.r.t it.

(1) 'X' can be represented as $\frac{'Z'}{2}$

(2) The value of 'Z' remains unchanged during competitive inhibition.

(3) The value of K_m decreases in the presence of a competitive inhibitor.

(4) The reaction ultimately reaches 'Z', which is not exceeded by any further rise in the concentration of substrate.

192. The proteinaceous molecule that joins a co-factor to form a holoenzyme is called

(1) Co-enzyme (2) Prosthetic group

(3) Inhibitor (4) Apoenzyme

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193. Read the given statements carefully.

Statement A: All amino acids are essential amino acids for the human body as all of them have to be obtained through our food.

Statement B: Tyrosine and phenylalanine are examples of aromatic amino acid.

Choose the **correct** option.

(1) Both statements A and B are incorrect
(2) Both statements A and B are correct
(3) Only statement A is incorrect
(4) Only statement B is incorrect

194. Match the following and choose the **correct** option.

a.	Acidic amino acid	(i)	Valine
b.	Basic amino acid	(ii)	Glutamic acid
c.	Neutral amino acid	(iii)	Lysine

(1) a(i), b(ii), c(iii)
(2) a(ii), b(iii), c(i)
(3) a(iii), b(ii), c(i)
(4) a(ii), b(i), c(iii)

195. How many compounds given in the box below will be found in the acid-soluble fraction upon chemical analysis of a living tissue?

DNA, Valine, Fructose, Cellulose, Insulin

Choose the **correct** option.

(1) Three
(2) Four
(3) Two
(4) Five

196. **Assertion (A) :** Frogs show indirect development.

Reason (R) : A part of their life cycle is spent in the water.

In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) (A) is true; (R) is false
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) Both (A) and (R) are false

197. If an adult frog is swimming inside a lake, then the mode of respiration that it exhibits will be

(1) Buccopharyngeal
(2) Pulmonary
(3) Cutaneous
(4) Branchial

198. Given below are the names of some structures:

(A) Alary muscles
(B) Phallic gland
(C) Spermatheca
(D) Ejaculatory duct
(E) Phallomere
(F) Collaterial glands

How many of the above mentioned structures are associated with the male reproductive system of cockroaches?

(1) Six
(2) Four
(3) Five
(4) Three

199. Select the **incorrect** statement w.r.t cartilage.

(1) The intercellular material of cartilage is composed of chondroitin sulfate which makes it solid and pliable.
(2) Intervertebral discs are made up of cartilage.
(3) Most of the cartilages present in the vertebrate embryo is replaced by bones in adults.
(4) Fibroblasts and collagen fibres are oriented in a definite pattern in its matrix.

200. Choose the **incorrect** statement w.r.t lipids.

(1) Lipids are generally water insoluble.
(2) Trihydroxy propane is a conjugated lipid.
(3) A diglyceride is composed of one glycerol molecule and two fatty acids.
(4) The neural tissues have complex lipids like sphingolipids and cholesterol.

❑ ❑ ❑

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